

Matgeo Presentation - Problem 4.3.38

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Problem Statement

Find the equation of the line joining the points $(3, 1)$ and $(9, 3)$.

Solution:

Solution :

Given

$$\mathbf{A} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \mathbf{B} = \begin{pmatrix} 9 \\ 3 \end{pmatrix} \quad (0.1)$$

Let us assume line equation to be:

$$\mathbf{n}^T \mathbf{x} = c \quad (0.2)$$

We get the line equation on solving

$$(\mathbf{A} \ \mathbf{B})^T \mathbf{n} = c \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The line passes through the points from (0.1) substituting, we get:

$$\begin{pmatrix} 3 & 9 \\ 1 & 3 \end{pmatrix}^T \mathbf{n} = c \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (0.3)$$

$$\begin{pmatrix} 3 & 1 \\ 9 & 3 \end{pmatrix} \mathbf{n} = c \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (0.4)$$

Solution:

Now by Gaussian Elimination solve:

$$\left(\begin{array}{cc|c} 3 & 1 & 1 \\ 9 & 3 & 1 \end{array} \right) \quad (0.5)$$

$$\begin{aligned} R_1 &\leftarrow \frac{1}{3}R_1 \\ \Rightarrow \left(\begin{array}{cc|c} 1 & \frac{1}{3} & \frac{1}{3} \\ 9 & 3 & 1 \end{array} \right) \end{aligned} \quad (0.6)$$

$$\begin{aligned} R_2 &\leftarrow R_2 - 9R_1 \\ \Rightarrow \left(\begin{array}{cc|c} 1 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & -2 \end{array} \right) \end{aligned} \quad (0.7)$$

By the assumption that line equation is $\mathbf{n}^T \mathbf{x} = 1$ which doesn't pass through origin we are not getting any solution. So our assumption is wrong and origin lies on the line. So consider

$$\mathbf{n}^T \mathbf{x} = 0 \quad (0.8)$$

Solution:

$c = 0$ because origin lies on the line and solving: so now, Assume the line equation:

$$\mathbf{n}^T \mathbf{x} = 0, \quad \mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix}$$

Line passes through points **A** and **B**

$$\mathbf{n}^T \mathbf{A} = 0 \implies 3n_1 + 1n_2 = 0 \quad (0.9)$$

$$\mathbf{n}^T \mathbf{B} = 0 \implies 9n_1 + 3n_2 = 0 \quad (0.10)$$

Matrix form:

$$\begin{pmatrix} 3 & 1 \\ 9 & 3 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.11)$$

Augmented matrix:

$$\left(\begin{array}{cc|c} 3 & 1 & 0 \\ 9 & 3 & 0 \end{array} \right) \quad (0.12)$$

Solution:

$$\begin{aligned} R_1 &\leftarrow \frac{1}{3}R_1 \\ \Rightarrow \left(\begin{array}{cc|c} 1 & \frac{1}{3} & 0 \\ 9 & 3 & 0 \end{array} \right) \end{aligned} \quad (0.13)$$

$$\begin{aligned} R_2 &\leftarrow R_2 - 9R_1 \\ \Rightarrow \left(\begin{array}{cc|c} 1 & \frac{1}{3} & 0 \\ 0 & 0 & 0 \end{array} \right) \end{aligned} \quad (0.14)$$

From first row:

$$n_1 + \frac{1}{3}n_2 = 0 \implies n_1 = -\frac{1}{3}n_2 \quad (0.15)$$

$$(0.16)$$

Solution:

Let,

$$n_2 = 3 \implies n_1 = -1 \quad (0.17)$$

$$\mathbf{n} = \begin{pmatrix} -1 \\ 3 \end{pmatrix} \quad (0.18)$$

$$\mathbf{n}^T \mathbf{x} = 0 \implies (-1 \quad 3) \mathbf{x} = 0 \quad (0.19)$$

Final Answer Required Equation is

$$\boxed{(-1 \quad 3) \mathbf{x} = 0}$$

C Source Code: points.c

```
#include <stdio.h>

// Function to compute normal vector for line through A(3,1) and B(9,3)
void get_normal_vector(double *n1, double *n2) {
    int x1 = 3, y1 = 1;
    int x2 = 9, y2 = 3;

    // Normal vector = [y2-y1, -(x2-x1)]
    *n1 = y2 - y1; // 3 - 1 = 2
    *n2 = -(x2 - x1); // -(9 - 3) = -6
}

int main() {
    double n1, n2;
    get_normal_vector(&n1, &n2);
    printf("Normal vector: n1=%lf, n2=%lf\n", n1, n2);
    return 0;
}
```

Python Script: call c.py

```
import ctypes

# Load C shared library
lib = ctypes.CDLL("./points.so")

# Prepare ctypes doubles
n1 = ctypes.c_double()
n2 = ctypes.c_double()

# Call C function
lib.get_normal_vector(ctypes.byref(n1), ctypes.byref(n2))
print("Normal_vector_from_C:", n1.value, n2.value)

# Save normal vector for plotting
normal_vector = (n1.value, n2.value)
```

Python Script: plot.py

```
import numpy as np
import matplotlib.pyplot as plt
import ctypes

# -----
# Load C library to get normal vector
# -----
lib = ctypes.CDLL("./points.so")
n1 = ctypes.c_double()
n2 = ctypes.c_double()
lib.get_normal_vector(ctypes.byref(n1), ctypes.byref(n2))
print("Normal vector from C:", n1.value, n2.value)

# -----
# Points A and B
# -----
A = np.array([3, 1])
B = np.array([9, 3])

# Direction vector along line
D = B - A

# Parameter t for plotting line
t = np.linspace(-1, 2, 100)
line_points = A[:, None] + D[:, None]*t
```

Python Script: plot.py

```
# -----  
# Plot line and points in 2D  
# -----  
plt.figure(figsize=(6,6))  
plt.plot(line_points[0], line_points[1], color='r', label='Line through A and B')  
plt.scatter([A[0], B[0]], [A[1], B[1]], color='b', s=50, label='Points A and B')  
  
# Optional: plot normal vector from origin  
origin = np.array([0,0])  
plt.quiver(*origin, n1.value, n2.value, angles='xy', scale_units='xy', scale=1,  
          color='g', label='Normal vector')  
  
plt.xlabel('X')  
plt.ylabel('Y')  
plt.title('Line and Normal Vector for Points (3,1) & (9,3)')  
plt.grid(True)  
plt.axis('equal')  
plt.legend()  
plt.savefig("line_normal_2d.png")  
plt.show()
```

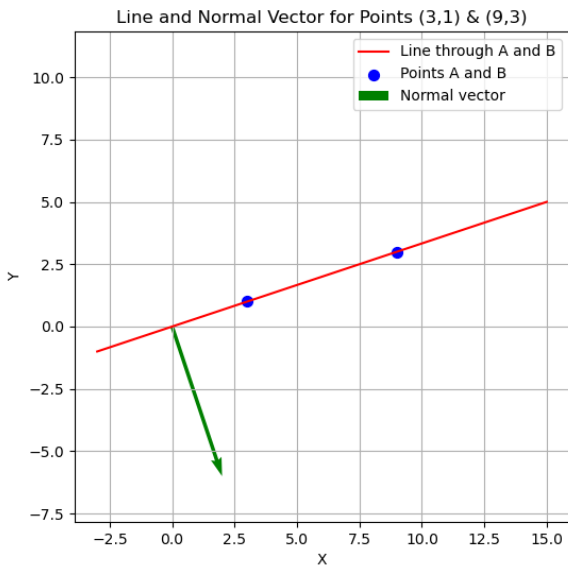


Figure: Plot for the unit vector along PQ